

# VINAYAK GUPTA

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## EDUCATION

### Birla Institute of Technology and Science, Pilani

MSc (Hons) in Biological Sciences | BE (Hons) in Manufacturing Engineering

2022 – Present

## EXPERIENCES

### Research Intern - Nanyang Technological University

Aug '26 – Present

Advisor: Professor Xueming Dong

Singapore

#### Hydrogen-Deuterium Exchange Epitope-omics Study on Intact Dengue Virus

- Analyzing Hydrogen-Deuterium Exchange Mass Spectrometry(HDX-MS) datasets using HDX-Workbench and perform manual kinetic calculations with HX-Express to characterize protein deuterium uptake.
- Performing mammalian cell culture using Baby Hamster Kidney(BHK-21) cells, including passaging, seeding, fixation, staining and plaque assays for DENV2 propagation and infection experiments.

### Research Intern - CSIR-Institute of Genomics and Integrative Biology

Feb '26 – Jul '26

Advisor: Professor Lipi Thukral

Delhi, India

#### Molecular Dynamics Simulations for RNA-Lipid Interactions

- Designed and executed molecular dynamics simulations of RNA molecules using GROMACS, including system preparation, solvation, ion addition, equilibration, production simulations and trajectory analysis.
  - Investigated RNA structural stability and conformational dynamics through trajectory analysis and interpretation of simulation-derived structural parameters, such as RMSD, RMSF, radius of gyration.
- #### Reproduction and Extension of the CplxCavity RNA Binding Site Prediction Framework
- Reconstructed a literature-derived computational pipeline for RNA cavity detection, integrating structural geometry analysis and extraction of physicochemical and geometric cavity descriptors.
  - Evaluated machine learning models such as XGBoost, for prediction of RNA ligand binding pockets.

## PUBLICATIONS

### Acoustic-assisted HDX-MS for Intact Virion Analysis

Status

Li Ching Pang, Ying Deng, **Vinayak Gupta**, Wen Zheng Lo, Shuangshuang Zhu, Xueming Dong

- Analyzed HDX-MS data from intact DENV2 virions to assess sequence coverage and identify antibody-associated changes in deuterium uptake.
- Interpreted protected regions following 2D22 binding and compared HDX-MS findings with the known cryo-EM-defined antibody interface.

## PROJECTS

### Antimicrobial Resistance(AMR) Prediction for Whole Genome Sequences using ML

Advisor: Professor Sandhya Marathe & Professor Tanmaya Mahapatra

Aug '25 – Dec '25

- Curated AMR datasets across multiple drug-pathogen pairs for machine learning model development.
- Integrated genomic features and MIC values to improve accuracy and generalization in prediction models.

### CNN-Based Model for Pneumonia Classification Using Chest CT Data

Dec '24

- Built a deep learning classification model to differentiate between healthy and pneumonia-affected lungs.
- Implemented image preprocessing pipelines using OpenCV and developed models in PyTorch for training.

### Bioinformatics Course Project

Advisor: Professor Shibasish Chowdhury

Jan '25 – May '25

- Predicted the structural(introns, exons, Promoter regions) and coding regions using GENSCAN.
- Created phylogenetic trees of six mammalian species using MEGA11, to find maximum parsimony.
- Pairwise gene alignment of multiple gene sequences using Basic Local Alignment Search Tool(BLASTn).

### Independent Study: Beta-Glucans and their Applications in Human Health and Plants

Advisor: Professor Balakumaran Chandrasekar

Jan '24 – May '24

- Conducted a literature review on the biological and agricultural applications of beta-glucans.

- Explored potential use cases in immunomodulation, disease prevention and crop protection.

### **Applied Biology Project Lab**

Advisor: Professor Shilpi Garg

Jan '24 – May '24

- Conducted a series of experiments to culture microorganisms, such as *E. coli*, to learn wet lab practices.
- Conducted assays including MICP, MIC, Beta-galactosidase activity and MTT cell viability.

### **SKILLS & TECHNICAL PROFICIENCY**

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**Languages:** Python[Fast.ai, Numpy, Pandas, Biopython, Scikit-Learn], C++, C, Git, HTML, bash

**Bioinformatics:** PyMol, Transcriptomic Data Analysis, HDX Workbench, GROMACS, ChimeraX

**Wet Lab:** Mammalian cell culture of Baby Hamster Kidney cell line(BHK-21)(plaque assay, passaging, fixation & staining, infecting), MIC cell viability assay, MTT and MICP assays, Beta-galactosidase assay.

**Design & Simulation:** AutoCAD, Fusion 360, Autodesk Maya, Arduino

### **COURSEWORK**

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**Computational Biology:** Biomolecular modeling, Bioinformatics, Recombinant DNA Technologies, Computer programming, Probability & Statistics, Machine Drawing, Engineering graphics

**Biological Sciences:** Developmental Biology, Immunology, Biological Chemistry, Genetics, Biophysics, Microbiology, Ecology & Environmental Sciences, Animal & Plant Physiology, Cell Biology